Clean Derivations for Sections 2.15–2.17

The Calculus of Selfishness (Karl Sigmund)

Preface and notation. These notes are rewritten in LaTeXby AI from scanned handwritten notes. So be careful while reading ahead. The equation numbering is kept aligned with Sigmund's Section 2.15–2.17, so that the steps can be checked directly against the text. The main state variable is $i \in \{0, 1, \ldots, M\}$, the number of individuals using strategy e_1 in a population of size M; consequently $M - i$ individuals use e_2.

1 [2.15] Stochastic processes and fixation probabilities

1.1 The birth–death recursion

Let p_{ij} be the probability that one update moves the population from state i to state j. Only nearest-neighbour moves (and no change) are possible, so write

$$p_{i,i+1} = b_i, \qquad p_{i,i-1} = d_i,$$

with $b_i + d_i \leq 1$. The states 0 and M are absorbing.

Let p_i be the probability that, starting from state i, the process eventually reaches M (fixation of e_1). Condition on the first update. There are three possibilities:

- the process moves to $i - 1$, with probability d_i;
- it stays at i, with probability $1 - b_i - d_i$;
- it moves to $i + 1$, with probability b_i.

Therefore, for $i = 1, \ldots, M - 1$,

$$p_i = d_i p_{i-1} + (1 - b_i - d_i) p_i + b_i p_{i+1}. \tag{2.62}$$

The boundary conditions are

$$p_0 = 0, \qquad p_M = 1.$$

Define the increments

$$y_i := p_i - p_{i-1}, \qquad i = 1, \ldots, M.$$

Rearranging (2.62),

$$\begin{aligned} 0 &= d_i(p_{i-1} - p_i) + b_i(p_{i+1} - p_i) \\ &= -d_i y_i + b_i y_{i+1}, \end{aligned}$$

so

$$y_{i+1} = \frac{d_i}{b_i} y_i. \tag{2.63}$$

1.2 Solving the recursion

Iterating (2.63) gives

$$y_2 = y_1 \frac{d_1}{b_1}, \qquad y_3 = y_1 \frac{d_1 d_2}{b_1 b_2}, \qquad y_k = y_1 \prod_{m=1}^{k-1} \frac{d_m}{b_m}.$$

Since $p_0 = 0$ manipulating the definition of y_1 gives,

$$p_i = \sum_{k=1}^{i} y_k.$$

In particular, $p_M = 1$ gives

$$1 = y_1 \sum_{j=0}^{M-1} \prod_{k=1}^{j} \frac{d_k}{b_k},$$

where an empty product (the $j = 0$ term) is 1. Hence

$$y_1 = \left[1 + \sum_{j=1}^{M-1} \prod_{k=1}^{j} \frac{d_k}{b_k} \right]^{-1}.$$

Therefore

$$\begin{aligned} p_i =& \sum_{k=1}^{i} y_k \\ =& \sum_{j=1}^{i-1} \left[y_1 \prod_{k=1}^{j} \frac{d_k}{b_k} \right] \\ =& y_1 \left[1 + \sum_{j=1}^{i-1} \prod_{k=1}^{j} \frac{d_k}{b_k} \right] \\ =& \frac{1 + \sum_{j=1}^{i-1} \prod_{k=1}^{j} \frac{d_k}{b_k}}{1 + \sum_{j=1}^{M-1} \prod_{k=1}^{j} \frac{d_k}{b_k}}. \end{aligned} \tag{2.65}$$

For a single e_1 mutant in an otherwise e_2 population, $i = 1$, so the numerator is 1. Thus the fixation probability is

$$\rho_{1,2} = p_1 = \left[1 + \sum_{j=1}^{M-1} \prod_{k=1}^{j} \frac{d_k}{b_k} \right]^{-1}. \tag{2.66}$$

1.3 Moran process with constant fitness ratio

Assume first that every e_1 individual has fitness r and every e_2 individual has fitness 1. In state i, the probability that the updating individual is an e_1 is i/M. For the population to move from i to $i-1$, that individual must then copy an e_2 model. Because model choice is proportional to fitness,

$$\Pr(\text{model is } e_2) = \frac{M - i}{ri + (M - i)}.$$

Therefore

$$d_i = \frac{i}{M}\frac{M-i}{ri+M-i}. \tag{2.67}$$

Similarly, to move from i to $i+1$, an e_2 individual must copy an e_1 model:

$$b_i = \frac{M-i}{M}\frac{ri}{ri+M-i}. \tag{2.68}$$

Taking the ratio, all state-dependent factors cancel:

$$\frac{d_i}{b_i} = \frac{1}{r}.$$

Substitution into (2.66) gives

$$\rho_{1,2} = \frac{1}{1+\sum_{j=1}^{M-1}\left(\frac{1}{r}\right)^j}.$$

Using the finite geometric sum,

$$1 + x + \cdots + x^{M-1} = \frac{1-x^M}{1-x}, \qquad x \neq 1,$$

with $x = r^{-1}$,

$$\rho_{1,2} = \frac{1-r^{-1}}{1-r^{-M}}, \qquad r \neq 1. \tag{2.69}$$

At $r = 1$, the mutant is neutral. Every term in the denominator of (2.66) is then 1, so

$$\rho_{1,2} = \frac{1}{M}.$$

This is the expected neutral fixation probability.

2 [2.16] Games in finite populations

2.1 Expected payoffs in state i

Let the payoff matrix be

$$A = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}. \tag{2.70}$$

There are i players of type e_1 and $M-i$ of type e_2. Players do not play against themselves.

For an e_1 player, there are $i-1$ possible e_1 opponents and $M-i$ possible e_2 opponents, out of $M-1$ possible opponents. Hence

$$F_i = \alpha\frac{i-1}{M-1} + \beta\frac{M-i}{M-1}. \tag{2.71}$$

For an e_2 player, there are i possible e_1 opponents and $M-i-1$ possible e_2 opponents, so

$$G_i = \gamma\frac{i}{M-1} + \delta\frac{M-i-1}{M-1}. \tag{2.72}$$

2.2 From payoff to fitness

Let $s \in [0,1]$ measure the strength of selection. Normalize the common baseline fitness to 1. The fitnesses are therefore the convex combinations

$$f_i = (1-s) + sF_i, \tag{2.73}$$
$$g_i = (1-s) + sG_i. \tag{2.74}$$

When $s = 0$, payoffs do not affect imitation; when $s = 1$, fitness is determined entirely by the game payoff.

2.3 Birth and death rates

A transition $i \to i+1$ occurs when an e_2 individual is selected for updating and copies an e_1 model. Thus

$$b_i = \frac{M-i}{M} \frac{if_i}{if_i + (M-i)g_i}. \tag{2.75}$$

Likewise, $i \to i-1$ occurs when an e_1 individual updates and copies an e_2 model:

$$d_i = \frac{i}{M} \frac{(M-i)g_i}{if_i + (M-i)g_i}. \tag{2.76}$$

The common factors cancel in the ratio:

$$\begin{aligned} \frac{d_i}{b_i} &= \frac{g_i}{f_i} \\ &= \frac{(1-s) + sG_i}{(1-s) + sF_i} \\ &= \frac{1 - s(1-G_i)}{1 - s(1-F_i)}. \end{aligned}$$

Hence

$$\frac{d_i}{b_i} = \frac{1 - s(1-G_i)}{1 - s(1-F_i)}. \tag{2.77}$$

Substituting (2.77) into the general fixation formula (2.66) yields

$$\rho_{1,2} = \left[1 + \sum_{j=1}^{M-1} \prod_{i=1}^{j} \frac{1 - s + sG_i}{1 - s + sF_i} \right]^{-1}. \tag{2.78}$$

3 [2.17] Limiting cases

3.1 Weak selection: first-order expansion in s

For $s \ll 1$, equation (2.77) becomes, to first order in s,

$$\begin{aligned} \frac{d_i}{b_i} &= \frac{1 - s(1-G_i)}{1 - s(1-F_i)} \\ &= \big[1 - s(1-G_i)\big]\big[1 + s(1-F_i) + O(s^2)\big] \\ &= 1 - s(F_i - G_i) + O(s^2). \end{aligned}$$

Define

$$H_i := F_i - G_i.$$

Then

$$\frac{d_i}{b_i} = 1 - sH_i + O(s^2). \tag{2.79}$$

Using (2.71) and (2.72),

$$\begin{aligned} H_i &= \frac{1}{M-1}\left[\alpha(i-1) + \beta(M-i) - \gamma i - \delta(M-i-1)\right] \\ &= \frac{\bar{e} + i\bar{f}}{M-1}, \end{aligned}$$

where it is convenient to define

$$\bar{e} = -\alpha + \beta M - \delta M + \delta, \tag{1}$$

$$\bar{f} = \alpha - \beta - \gamma + \delta. \tag{2.80}$$

Thus, to first order in s,

$$\rho_{1,2} = \left[1 + \sum_{k=1}^{M-1}\prod_{i=1}^{k}(1 - sH_i)\right]^{-1} + O(s^2). \tag{2.81}$$

3.2 Reducing the double sum

For each k,

$$\prod_{i=1}^{k}(1 - sH_i) = 1 - s\sum_{i=1}^{k} H_i + O(s^2),$$

so

$$\sum_{k=1}^{M-1}\prod_{i=1}^{k}(1 - sH_i) = M - 1 - s\sum_{k=1}^{M-1}\sum_{i=1}^{k} H_i + O(s^2).$$

In the double sum, H_i appears once for every $k \geq i$, namely for $k = i, i+1, \ldots, M-1$. There are $M - i$ such values of k. Therefore

$$\sum_{k=1}^{M-1}\prod_{i=1}^{k}(1 - sH_i) = M - 1 - s\sum_{i=1}^{M-1}(M-i)H_i + O(s^2). \tag{2.82}$$

It remains to evaluate

$$S := \sum_{i=1}^{M-1}(M-i)H_i = \frac{1}{M-1}\sum_{i=1}^{M-1}(M-i)(\bar{e} + i\bar{f}).$$

Split the sum:

$$S = \frac{1}{M-1}\left[\bar{e}\sum_{i=1}^{M-1}(M-i) + \bar{f}\sum_{i=1}^{M-1} i(M-i)\right].$$

The first sum is

$$\sum_{i=1}^{M-1}(M-i) = \frac{M(M-1)}{2}.$$

For the second,

$$\begin{aligned}\sum_{i=1}^{M-1} i(M-i) &= M\sum_{i=1}^{M-1} i - \sum_{i=1}^{M-1} i^2 \\ &= M\frac{M(M-1)}{2} - \frac{M(M-1)(2M-1)}{6} \\ &= \frac{M(M-1)(M+1)}{6}.\end{aligned}$$

Hence

$$S = \frac{M}{6}\big(3\bar{e} + (M+1)\bar{f}\big). \tag{2}$$

It is useful to rewrite the bracket in terms of the original payoff entries. Define

$$e := \alpha + 2\beta - \gamma - 2\delta, \tag{3}$$

$$f := 2\alpha + \beta + \gamma - 4\delta. \tag{2.84-def}$$

A direct substitution of $\bar{e}$ and $\bar{f}$ gives

$$3\bar{e} + (M+1)\bar{f} = eM - f.$$

Therefore

$$\sum_{i=1}^{M-1} (M-i)H_i = \frac{M}{6}(eM - f). \tag{2.84}$$

3.3 Weak-selection fixation probability

Using (2.82) and (2.84) in (2.81), the denominator becomes

$$M - s\frac{M(M-1)}{6}(eM - f) + O(s^2).$$

Factoring out M gives

$$\begin{aligned}\rho_{1,2} &= \frac{1}{M\left[1 - \frac{s}{6}(eM-f)\right]} + O(s^2) \\ &= \frac{1}{M}\left[1 - \frac{s}{6}(eM-f)\right]^{-1} + O(s^2).\end{aligned}$$

Thus

$$\rho_{1,2} = \frac{1}{M}\left[1 - \frac{s}{6}(eM-f)\right]^{-1} + O(s^2). \tag{2.85}$$

3.4 When is e_1 advantageous?

The neutral fixation probability is $1/M$. Therefore e_1 is advantageous precisely when

$$\rho_{1,2} > \frac{1}{M}.$$

For sufficiently small positive s, (2.85) implies

$$\left[1 - \frac{s}{6}(eM-f)\right]^{-1} > 1,$$

which is equivalent to

$$eM - f > 0.$$

Thus

$$eM > f. \tag{2.86a}$$

Substituting the definitions of e and f gives

$$M(\alpha + 2\beta - \gamma - 2\delta) > 2\alpha + \beta + \gamma - 4\delta,$$

or, after collecting coefficients,

$$\alpha(M-2) + \beta(2M-1) > \gamma(M+1) + \delta(2M-4). \tag{2.86}$$

3.5 Large-population limit

Divide (2.86) by M and let $M \to \infty$. The terms of order $1/M$ vanish, leaving

$$\alpha + 2\beta > \gamma + 2\delta. \tag{2.87}$$

To connect this with the usual two-strategy replicator notation, subtract the diagonal entries from the corresponding columns of A. This does not change the replicator dynamics and transforms (2.70) into

$$\begin{pmatrix} 0 & a \\ b & 0 \end{pmatrix}, \qquad a = \beta - \delta, \quad b = \gamma - \alpha.$$

Then (2.87) becomes

$$b < 2a.$$

3.6 Interpretation in the three two-strategy cases

With the normalized matrix

$$A = \begin{pmatrix} 0 & a \\ b & 0 \end{pmatrix},$$

let x be the frequency of e_1. The payoff difference is

$$F_1 - F_2 = a(1-x) - bx = a - (a+b)x,$$

so the replicator equation is

$$\dot{x} = x(1-x)\big[a - (a+b)x\big]. \tag{2.87'}$$

If a and b are both positive, the interior equilibrium is

$$\hat{x} = \frac{a}{a+b},$$

and it is stable. The condition $b < 2a$ is then equivalent to

$$a + b < 3a \quad \Longleftrightarrow \quad \hat{x} > \frac{1}{3}.$$

Thus, in the stable-coexistence case, e_1 is advantageous when its equilibrium frequency exceeds $1/3$. Symmetrically, e_2 is advantageous when $\hat{x} < 2/3$; consequently, if $1/3 < \hat{x} < 2/3$, both strategies can be advantageous.

If $a < 0$ and $b < 0$, the interior equilibrium is unstable (bistability). The same condition $\hat{x} > 1/3$ identifies when e_1 is advantageous. In this case that condition also means that the basin of attraction of e_1 is larger than that of e_2; hence an advantageous strategy is risk-dominant in the large-population limit.

If e_1 dominates e_2 ($a \geq 0$, $b \leq 0$, with at least one strict inequality), then $b < 2a$ automatically holds. Thus a dominant strategy is always advantageous.

3.7 Comparing the two fixation probabilities

The corresponding weak-selection calculation for a single e_2 mutant gives

$$\rho_{2,1} = \frac{1}{M}\left[1 - \frac{s}{6}(e_2 M - f_2)\right]^{-1} + O(s^2),$$

where

$$e_2 = -2\alpha - \beta + 2\gamma + \delta, \qquad f_2 = -4\alpha + \beta + \gamma + 2\delta.$$

Comparing the two fixation probabilities yields, after cancellation and simplification,

$$\rho_{1,2} > \rho_{2,1} \quad \Longleftrightarrow \quad (M-2)(\alpha - \delta) > M(\gamma - \beta). \tag{2.88}$$

As $M \to \infty$, this becomes

$$\alpha - \delta > \gamma - \beta,$$

which is exactly the risk-dominance condition $a > b$ in the normalized matrix.

3.8 Rare innovation and the two-state stationary distribution

The deterministic imitation process has absorbing homogeneous states. If a small innovation probability $\mu > 0$ is added, a homogeneous population can occasionally produce a single mutant, after which imitation either eliminates or fixes that mutant. When innovation is sufficiently rare, these two processes occur on well-separated time scales, so the long-run dynamics can be approximated by a Markov chain whose states are the two homogeneous populations:

$$\text{state 1: all } e_1, \qquad \text{state 2: all } e_2.$$

The transition matrix is

$$P = \begin{pmatrix} 1 - \rho_{2,1} & \rho_{2,1} \\ \rho_{1,2} & 1 - \rho_{1,2} \end{pmatrix}. \tag{2.89}$$

Let the stationary distribution be $\pi = (\pi_1, \pi_2)$. It satisfies

$$\pi P = \pi, \qquad \pi_1 + \pi_2 = 1.$$

The first component gives

$$\pi_1(1 - \rho_{2,1}) + \pi_2 \rho_{1,2} = \pi_1,$$

so

$$\pi_1 \rho_{2,1} = \pi_2 \rho_{1,2}.$$

Hence

$$\frac{\pi_1}{\pi_2} = \frac{\rho_{1,2}}{\rho_{2,1}}.$$

Using $\pi_1 + \pi_2 = 1$,

$$\pi_1 = \frac{\rho_{1,2}}{\rho_{1,2} + \rho_{2,1}},$$
$$\pi_2 = \frac{\rho_{2,1}}{\rho_{1,2} + \rho_{2,1}}.$$

Therefore

$$\pi = \left(\frac{\rho_{1,2}}{\rho_{1,2} + \rho_{2,1}}, \frac{\rho_{2,1}}{\rho_{1,2} + \rho_{2,1}}\right). \tag{2.90}$$

In particular, in a bistable game and for large M, the risk-dominant strategy is precisely the one whose homogeneous state receives the larger stationary weight.